

Music & Mood Analysis Using Spotify Data

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Introduction

- Music is something people interact with emotionally every day. Different songs can make us feel happy, energetic, calm, emotional, or even nostalgic.
- While emotions in music are usually considered subjective, modern music platforms such as Spotify now provide measurable audio features that allow us to statistically analyze musical characteristics.

Why we chose this topic

- We were interested in understanding how streaming platforms use audio features to create personalized recommendations and mood-based playlists.
- Understanding relationships between musical features and emotional tone has practical applications in areas such as music recommendation systems, playlist generation, listener behavior analysis, and mood-based music streaming.

Objective

The primary objective of this study was to investigate whether measurable musical attributes can explain variations in the emotional tone of songs.

We focused on audio features such as tempo, energy, and danceability, and analyzed their relationship with valence, which represents the emotional positivity of a song.

To study these relationships, we applied statistical techniques including exploratory data analysis, hypothesis testing, correlation analysis, and multiple linear regression.

The broader aim of the project was to understand whether emotional patterns in music can be quantitatively analyzed using numerical audio features.

Research Question

The central research question of our study was whether measurable musical characteristics can significantly explain variations in the emotional tone of songs.

This is important in applications such as recommendation systems, playlist generation, and mood-based music analysis.

Dataset Description

Dataset Source:

- Spotify Tracks Dataset obtained from Kaggle

Dataset Information:

- Approximately 114000 songs
- 21 variables
- Each row represents a song
- Each column represents an audio feature or metadata attribute

Selected Variables:

- Tempo
- Energy
- Danceability
- Valence

How the features are generated

- **The audio features in the dataset are generated using Spotify's proprietary audio analysis system, which uses machine learning and digital signal processing techniques to analyze raw audio signals.**
- **For tempo:** Beat-tracking algorithms detect repeating rhythmic patterns and calculate the number of beats occurring per minute
- **For Danceability:** Spotify analyzes - rhythm stability, beat, regularity, tempo consistency, groove patterns Songs with: steady rhythm, predictable beats, strong groove receive higher danceability scores.

How the features are generated

- **For valence:** Valence is estimated using machine learning models trained on musical characteristics associated with emotional tone. Spotify's system looks at:
harmony, chord progression, brightness, rhythm patterns, tonal characteristics Then predicts whether a song sounds emotionally positive or negative.
- Spotify is NOT understanding human emotions. It is identifying mathematical sound patterns commonly associated with emotional tone.

Key Variables

Among the 21 variables available in the dataset, we selected **four key audio features** that are closely related to musical structure and emotional perception.

- **TEMPO**: Tempo represents the speed of a song and is measured in beats per minute, or BPM.
- **ENERGY**: Energy measures the intensity and activity level of a song.
- **DANCEABILITY**: Danceability represents how suitable a song is for dancing.
- **VALENCE**: Valence represents the emotional positivity of a song.

Data Preprocessing

STEPS:

- Loaded dataset into R
- Removed missing values
- Selected relevant variables
- Verified numeric data types
- Created cleaned dataset for analysis

- **Removing missing values:**

```
music <- na.omit(music)
```

Missing values were removed using the `na.omit()` function to ensure that incomplete observations did not affect statistical analysis

- **Check for missing values:**

```
colSums(is.na(music))
```

After cleaning, missing values were checked again to verify that the dataset contained no incomplete observations.

- **Choosing relevant variables:**

```
data <- music[, c("tempo", "energy", "danceability", "valence")]
```

Only variables directly relevant to the research question were selected in order to simplify analysis and improve interpretability

- **Verifying data types of variables:** The selected variables were verified to ensure they were numerical and suitable for statistical techniques such as correlation analysis and regression.

Exploratory Data Analysis

Purpose of EDA

- To understand the distribution and behavior of musical variables
- To identify patterns, relationships, and variability within the dataset
- To examine how audio features relate to song valence (mood positivity)
- To support hypothesis testing and regression modelling

EDA Techniques Used

- Summary statistics
- Histograms
- Scatter plots
- Correlation matrix
- Side-by-side boxplots

Summary Statistics

Summary statistics were used to examine the distribution and variability of the selected musical features.

Output

R Code

```
summary(data)
```

	tempo	valence	energy	danceability
Min. :	0.00	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:	99.22	1st Qu.:0.2600	1st Qu.:0.4720	1st Qu.:0.4560
Median :	122.02	Median :0.4640	Median :0.6850	Median :0.5800
Mean :	122.15	Mean :0.4741	Mean :0.6414	Mean :0.5668
3rd Qu.:	140.07	3rd Qu.:0.6830	3rd Qu.:0.8540	3rd Qu.:0.6950
Max. :	243.37	Max. :0.9950	Max. :1.0000	Max. :0.9850

Key Observations

- Tempo showed the widest numerical range
- Energy and danceability were concentrated toward higher values
- Valence showed balanced variability suitable for modelling

Histograms of Musical Features

Histograms were used to examine the distribution, concentration, skewness, and variability of musical features, helping identify overall patterns in listener and song characteristics

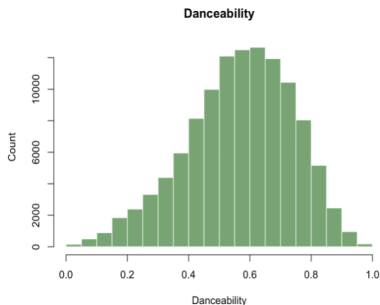


Figure: Danceability

The slightly left-skewed distribution shows that listeners prefer moderately to highly danceable music with uplifting rhythms.

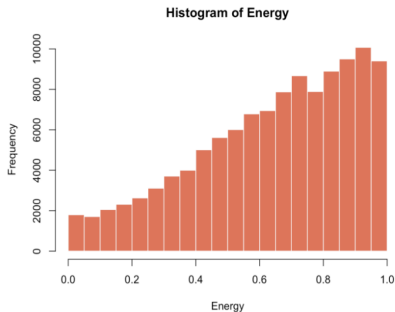


Figure: Energy

The left-skewed energy distribution indicates that high-energy and emotionally intense tracks dominate mainstream listening.

Histograms of Musical Features

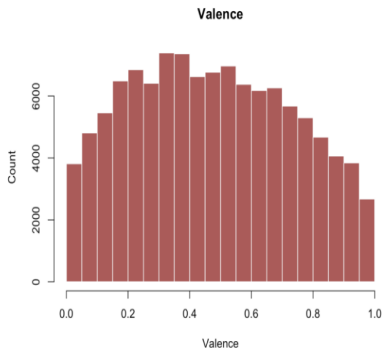


Figure: Valence

The moderately bell-shaped valence distribution suggests that neutral-to-positive moods are most common in the dataset.

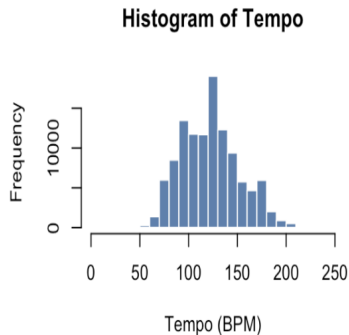


Figure: Tempo

The slightly right-skewed tempo distribution, concentrated between 80–160 BPM, reflects the pacing commonly found in popular music genres.

Scatter Plot Analysis

Scatter plots were used to examine relationships between musical features and valence. They help visualize correlation patterns and the strength of association between variables.

The upward trend indicates that higher danceability is associated with higher valence values.

Danceability appears to be the strongest predictor of song mood.

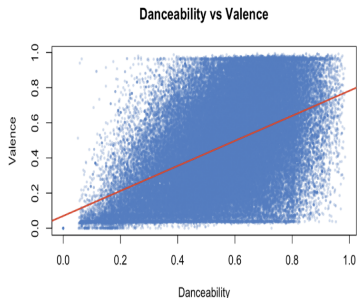


Figure: Danceability vs Valence

Scatter Plot Analysis

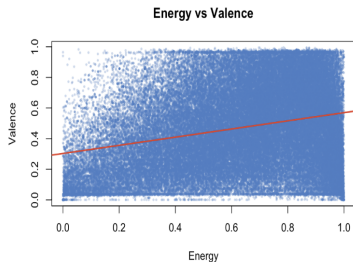


Figure: Energy vs Valence

The moderate positive correlation indicates that energetic music may contribute to more uplifting listener experiences.

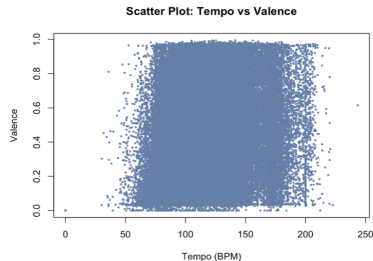


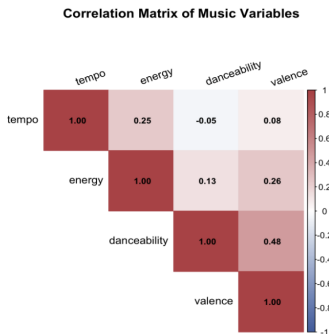
Figure: Tempo vs Valence

The weakly scattered relationship suggests that songs with similar tempos can still convey very different emotional moods.

Energy appears to influence emotional positivity more strongly than tempo

Correlation Matrix Analysis

A correlation matrix was used to measure the strength and direction of relationships between the selected variables. It helps identify which predictors are most strongly associated with valence before regression analysis.



Key Findings

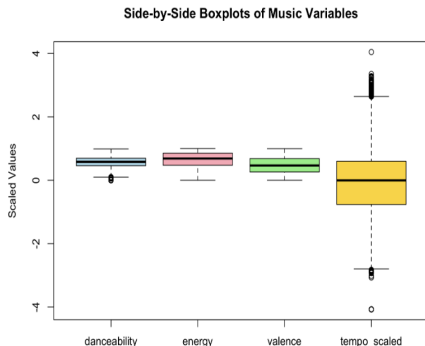
- Danceability has strongest correlation with valence ($r = 0.48$)
- Energy shows moderate positive relationship ($r = 0.26$)
- Low multicollinearity supports regression analysis

Danceability appears to be the strongest predictor of emotional positivity among the variables considered.

Figure: Correlation Matrix of Music Variables

Boxplot Analysis of Music Variables

Boxplots were used to compare the distribution, spread, and presence of outliers across danceability, energy, valence and tempo



Observations

- Tempo exhibits the highest variability and largest number of outliers
- Energy values are concentrated toward higher ranges
- Danceability and valence display relatively balanced distributions

Danceability and energy appear more stable predictors than tempo.

Figure: Side-by-Side Boxplots of Music Variables

Hypothesis Testing

A statistical method used to determine whether an observed relationship is statistically significant or could have occurred by random chance.

Purpose of Analysis

- During exploratory data analysis, danceability and energy showed relatively clear positive relationships with valence through scatter plots and correlation analysis
- This section investigates whether tempo has a statistically significant association with song valence using both parametric and non-parametric statistical tests.
- Basically **Do fast songs and slow songs differ in emotional positivity?**

Hypotheses

Null Hypothesis (H_0)

- There is no significant relationship between tempo and valence

$$H_0 : \mu_{\text{fast}} = \mu_{\text{slow}}$$

Alternative Hypothesis (H_1)

- Songs with lower tempo tend to exhibit lower valence

$$H_1 : \mu_{\text{fast}} \neq \mu_{\text{slow}}$$

SPLIT FAST AND SLOW SONGS

- Songs were divided into slow and fast groups using the median tempo value of 122.02 BPM. Songs below the median were classified as slow, while songs greater than or equal to the median were classified as fast.
- And then we compare average valence of fast songs VS average valence of slow songs

T - TEST

- A t-test compares whether the means of two groups are significantly different.
- Independent two-sample t-test performed

Hypothesis Testing Results

Dataset Information

- Number of songs analyzed: 114000
- Median tempo: 122.02 BPM

Results

- Mean valence of fast songs: 0.482
- Mean valence of slow songs: 0.466
- T-statistic = 10.225
- p-value < 0.05

Results

Fast songs mean valence:	0.482
Slow songs mean valence:	0.466
T-statistic:	10.225
p-value:	$< 2.2 \times 10^{-16}$

Figure: Result of t-test

Conclusion

- The null hypothesis is rejected due to the very small p-value.
- Tempo shows a statistically significant relationship with valence.

Interpretation of Hypothesis Testing

- Tempo and valence are indeed statistically associated
- Though difference was small since dataset massive (114,000 songs), even tiny differences become statistically significant.
- Although the relationship was statistically significant, the practical effect remained relatively weak, indicating that tempo alone is not a strong determinant of emotional tone.

Correlation Analysis

Correlation analysis was performed to measure the strength and direction of the linear relationship between tempo and valence. The correlation coefficient (r) helps determine whether changes in song tempo are associated with changes in emotional positivity.

- We performed Pearson correlation analysis in R.

```
correlation <- cor(datatempo, datavalence) print(correlation)
```

OUTPUT: $r=0.0783$

- Weak positive linear relationship between tempo and valence
- Faster songs tend to have slightly higher emotional positivity

MANN-WHITNEY U TEST

To validate the robustness of the hypothesis testing results, a Mann-Whitney U Test was additionally performed as a non-parametric alternative to the t-test. This makes it suitable for datasets that may exhibit skewness or non-normal behavior

- R code.

```
wilcox result <- wilcox.test(valence ~ tempo group, data = data) print(wilcox result)
```

OUTPUT: $W = 1.677 \times 10^9$

p-value $< 2.2 \times 10^{-16}$

- p-value < 0.05
- The Mann-Whitney test also produced a p-value significantly lower than 0.05, leading to rejection of the null hypothesis once again, confirming that the relationship remained statistically significant even under a non-parametric approach.

Regression Analysis

Regression analysis was performed to understand how musical characteristics are associated with the emotional positivity of songs, represented by valence. It helps quantify the relationship between audio features and mood, allowing us to identify which musical patterns are most strongly linked to emotionally positive listening experiences.

Response Variable

- Valence

Predictor Variables

- Tempo
- Energy
- Danceability

The analysis helps identify which musical features contribute most strongly to perceived song mood.

Multiple Linear Regression Model

Multiple Linear Regression was selected because song mood is influenced by several musical features together rather than a single variable alone.

Regression Equation

$$\hat{\text{Valence}} = \beta_0 + \beta_1(\text{Tempo}) + \beta_2(\text{Energy}) + \beta_3(\text{Danceability}) + \epsilon$$

Where

- β_0 = Intercept
- $\beta_1, \beta_2, \beta_3$ = Regression coefficients
- ϵ = Random error term

Assumptions of Multiple Linear Regression

The following assumptions were verified before fitting the regression model:

- **Linearity:** Predictors should show a linear relationship with valence
- **Independence of Errors:** Residuals should remain independent
- **Normality of Residuals:** Residuals should approximately follow a normal distribution
- **Homoscedasticity:** Residual variance should remain reasonably constant
- **Low Multicollinearity:** Predictors should not be highly correlated

Diagnostic plots and correlation analysis were used to validate these assumptions.

Regression Results

Fitted Regression Model

$$\hat{Valence} = -0.09199 + 0.000481(Tempo) + 0.1896(Energy) + 0.6804(Danceability)$$

Key Findings

- Tempo is statistically significant but has a relatively weak effect
- Energy positively contributes to emotional positivity
- Danceability shows the strongest positive contribution

All predictors showed statistical significance with p -values $< 2 \times 10^{-16}$.

Model Evaluation Metrics

R^2
0.2694

- Explains 26.94% of variation in valence
- Moderate explanatory power

Adjusted R^2
0.2694

- Predictors contribute meaningfully
- Minimal unnecessary complexity

RSE
0.2216

- Low prediction error
- Predictions remain reasonably close to actual values

The model captures meaningful relationships between musical features and emotional positivity

Overall Model Significance

F-Statistic

$$1.401 \times 10^4$$

- Model statistically significant overall
- Strong predictor relationship observed

p-value

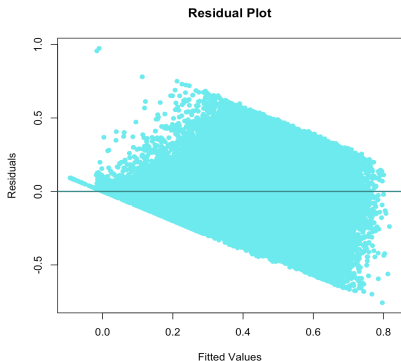
$$< 2.2 \times 10^{-16}$$

- Strong evidence against null hypothesis
- At least one predictor significantly affects valence

The regression model captures meaningful relationships between musical features and song mood.

Residual Plot Analysis

Residual plots were used to examine error patterns and verify assumptions of linearity and constant variance.



Observations

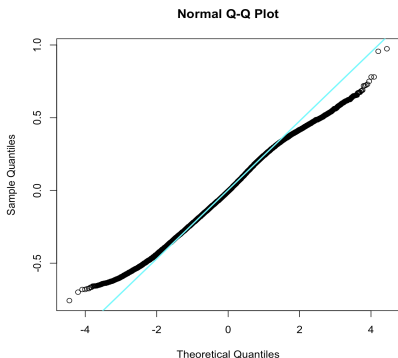
- Residuals are approximately centered around zero
- Slight heteroscedasticity is visible
- No strong nonlinear pattern is observed

Inference

- Linearity assumption is reasonably satisfied
- Residual variance remains reasonably stable

Normal Q-Q Plot Analysis

The Q-Q plot was used to assess whether regression residuals follow an approximately normal distribution.



Normal Q-Q Plot

Observations

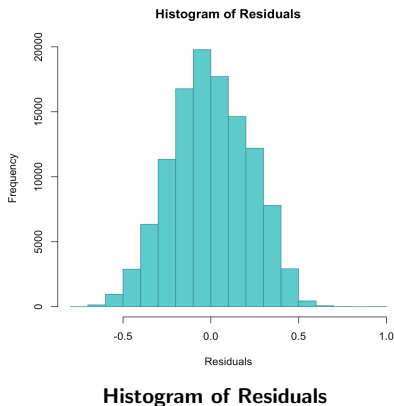
- Most points lie close to the reference line
- Minor deviations appear at the tails
- Residual distribution appears approximately normal

Inference

- Normality assumption is reasonably satisfied
- Extreme residual deviations remain limited

Histogram of Residuals

A histogram of residuals was used to visually examine the distribution of regression errors.



Observations

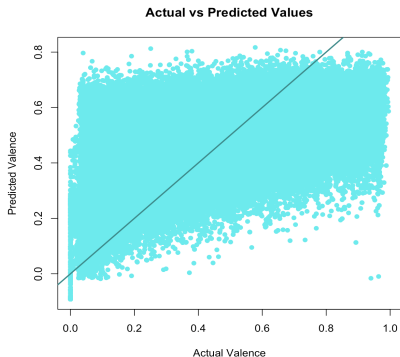
- Residuals follow an approximately normal distribution
- Residuals remain fairly symmetric
- Most residuals cluster near zero

Inference

- Error terms are reasonably normally distributed
- Regression assumptions are generally supported

Actual vs Predicted Analysis

This plot compares observed valence values with regression model predictions.



Actual vs Predicted Plot

Observations

- Predicted values generally follow observed trends
- Moderate prediction variability is still present
- The model captures overall mood patterns reasonably well

Inference

- Regression model provides a reasonable fit
- Predictions align reasonably with observed valence values

Overall Regression Conclusions

The regression model showed that musical features contribute meaningfully to predicting emotional positivity in songs.

Key Conclusions

- **Danceability** showed the strongest positive relationship with valence, indicating that rhythmically engaging songs are generally perceived as happier.
- **Energy** also contributed positively, suggesting that energetic tracks are commonly associated with uplifting listener experiences.
- **Tempo** showed only a weak influence on mood, implying that emotional perception depends on multiple musical characteristics.
- **Diagnostic analysis** supported that the regression model is statistically meaningful and reasonably reliable.

These findings highlight applications in music recommendation systems, playlist generation, and mood-based personalization.

Final Conclusion

- Statistical analysis showed that musical audio features contribute meaningfully to emotional tone represented by valence.
- Danceability emerged as the strongest predictor of emotional positivity, followed by energy and tempo.
- Tempo showed a statistically significant but weak association with valence, indicating that song speed alone does not strongly determine mood.
- Multiple Linear Regression demonstrated that measurable musical characteristics can partially explain variations in emotional tone.
- The study highlights the usefulness of statistical methods and large-scale music datasets in understanding patterns in musical mood and listener perception.

Challenges and Future Scope

Challenges

- Mood is approximated using valence rather than directly measured emotions
- Emotional interpretation of music remains subjective
- Lyrics and cultural context were not included

Future Scope

- Include additional audio features and metadata for more detailed mood analysis.
- Apply advanced machine learning techniques to improve mood prediction accuracy.
- Perform lyrical sentiment analysis to study how song lyrics contribute to emotional tone.
- Combine music datasets with personality trait datasets to analyze how musical preferences and listening behavior may relate to different personality types.

Thank You